

Due on monday, December 3rd.

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1. Potential scattering.

Consider a particle of mass  $m$  on a one-dimensional line. The potential is zero to the left of  $x = -a$  and has a constant positive value to the right of  $x = a$ . The potential has unknown features within the region  $x \in [-a, a]$ .

$$V(x) = \begin{cases} 0 & \text{for } x < -a \\ \text{unknown} & \text{for } -a < x < a \\ V_0 & \text{for } x > a \end{cases}$$

The Schroedinger equation for this system is found/known to have a solution with energy  $E$  with the following forms in the regions outside  $[-a, a]$ :

$$\begin{aligned} \psi(x) &= \alpha e^{ik_1 x} + \beta e^{-ik_1 x} && \text{for } x < -a \\ \psi(x) &= \gamma e^{ik_2 x} && \text{for } x > a \end{aligned}$$

where  $k_1$  and  $k_2$  are positive real constants, and

$$\alpha = \frac{k_1 + ik_2}{k_1 - ik_2} \quad ; \quad \beta = \frac{ik_1}{k_1 - ik_2} \quad ; \quad \gamma = \frac{\sqrt{k_1 k_2}}{k_1 - ik_2}$$

- (a) [4 pts.] Express  $k_1$  in terms of  $E$ .
- (b) [2 pts.] Is  $E$  larger or smaller than  $V_0$ ? Explain why.
- (c) [5 pts.] Express  $k_2$  in terms of  $E$  and  $V_0$ .
- (d) [3 pts.] If we want to interpret this as a scattering problem, which parts of the wavefunction would you interpret as incident wave, reflected wave, and transmitted wave?
- (e) [4 pts.] Calculate the reflection coefficient.
- (f) [5 pts.] Calculate the transmission coefficient.  
(Careful: remember momentum factors.)
- (g) [2 pts.] Check that the sum of the two coefficients makes sense.

2. Suppose  $\psi_0(x)$  is a normalized wavefunction with

$$\langle \hat{x} \rangle_{\psi_0} = \langle \psi_0 | \hat{x} | \psi_0 \rangle = x_0 \quad \langle \hat{p} \rangle_{\psi_0} = \langle \psi_0 | \hat{p} | \psi_0 \rangle = p_0$$

where  $x_0$  and  $p_0$  are constants. The subscript indicates the state in which the expectation values are being calculated. Define a new wavefunction

$$\psi_{new}(x) = e^{iqx/\hbar} \psi_0(x)$$

where  $q$  is a real quantity having dimensions of momentum.

- (a) [6 pts.] What is the expectation value  $\langle \hat{x} \rangle_{\psi_{new}}$  in the state  $\psi_{new}(x)$ ?
  - (b) [7 pts.] What is the expectation value  $\langle \hat{p} \rangle_{\psi_{new}}$  in the state  $\psi_{new}(x)$ ?
  - (c) [SELF] Based on your results, interpret in one sentence the physical significance of multiplying a wavefunction by the factor  $e^{iqx/\hbar}$ .
3. Consider a particle of mass  $m$  on a one-dimensional line. The state of the particle is described by wavefunction  $\psi(x)$ .

- (a) [0 pts.] Look up and write down correctly the definition of the probability current density.
- (b) [4 pts.] We write the wavefunction in amplitude-phase form:

$$\psi(x) = A(x)e^{i\theta(x)}$$

where both  $A(x)$  and  $\theta(x)$  are real functions.

Show that the probability current density is given by

$$j(x) = [A(x)]^2 \frac{\hbar}{m} \partial_x \theta(x)$$

i.e., the probability current is proportional to the *gradient* of the phase.

- (c) [2 pts.] Show that there is no current in a region where the wavefunction is real.
4. Consider the time-dependent Schroedinger equation (TDSE) for a single particle on a one-dimensional line:

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x) \psi(x, t)$$

The potential  $V(x)$  is real-valued, i.e.,  $V^*(x) = V(x)$ .

- (a) [**6 pts.**] Using the TDSE, show that the probability density  $\rho(x, t) = \psi^* \psi$  and the probability current density  $j(x, t)$  satisfy the one-dimensional continuity equation

$$\frac{\partial j}{\partial x} = - \frac{\partial \rho}{\partial t}$$

- (b) [**SELF**] Guess the form of the continuity equation for the probability density, for the case of a particle in three spatial dimensions.
- (c) [**SELF**] What does the continuity equation imply physically?  
The wikipedia page on “Continuity Equation” has a section called “quantum mechanics”. You might try reading through.